Assignment 2

Question 1. Bell States and Entanglement (18 points)

The four Bell state are maximally entangled two-qubit states:

$$|\phi^{\pm}\rangle = \frac{|00\rangle \pm |11\rangle}{\sqrt{2}}, \qquad |\psi^{\pm}\rangle = \frac{|01\rangle \pm |10\rangle}{\sqrt{2}}$$

a. **(4 points)** Write each of the four Bell states as column vectors in the computational basis using the ordering $|00\rangle, |01\rangle, |10\rangle, |11\rangle$.
Tensor products in matrix notation work as follows:

$$\begin{bmatrix} a \\ b \end{bmatrix} \otimes \begin{bmatrix} c \\ d \end{bmatrix} = \begin{bmatrix} a\begin{bmatrix} c \\ d \end{bmatrix} \\ b\begin{bmatrix} c \\ d \end{bmatrix} \end{bmatrix} = \begin{bmatrix} a \cdot c \\ a \cdot d \\ b \cdot c \\ b \cdot d \end{bmatrix}.$$

b. **(4 points)** The four Bell states are orthonormal. Prove this for $|\phi^{+}\rangle$, i.e., prove that it is normalized and that its inner product with the other three Bell states is zero.

c. **(2 points)** Consider the two-qubit state $|\Phi\rangle = \frac{1}{2}[|00\rangle + |01\rangle + |10\rangle - |11\rangle]$. Determine whether it is entangled or separable.

d. **(2 points)** Alice and Bob share the state $|\phi^{+}\rangle$. They each measure their qubit in $\hat{Z}$ basis. List all possible outcomes, their probabilities, and the values obtained by Alice and Bob in each case.

e. **(6 points)** Rewrite $|\phi^{+}\rangle$ in $\hat{X}$ basis $|++\rangle, |+-\rangle, |-+\rangle, |--\rangle$. Suppose Alice and Bob measure their qubit in $\hat{X}$ basis. List all possible outcomes, their probabilities, and the values obtained by Alice and Bob in each case.

Question 2. The GHZ State and Local Realism (17 points)

The Greenberger-Horne-Zeilinger (GHZ) state is a three-qubit entangled state:

$$|GHZ\rangle = \frac{|000\rangle + |111\rangle}{\sqrt{2}}.$$

For a multi-qubit state, operators can be decomposed such that constituents act individually:

$$\left(\hat{A} \otimes \hat{B} \otimes \hat{C}\right)(|a\rangle \otimes |b\rangle \otimes |c\rangle) = \hat{A}|a\rangle \otimes \hat{B}|b\rangle \otimes \hat{C}|c\rangle.$$

a. **(3 points)** Show that $|GHZ\rangle$ cannot be written as a product state $|\psi_A\rangle \otimes |\psi_B\rangle \otimes |\psi_C\rangle$.

b. **(2 points)** Alice, Bob and Charlie each measure qubits A, B and C respectively in $\hat{Z}$ basis. List all possible outcomes and their probabilities.

c. **(4 points)** Show that $\langle GHZ|\hat{X}_A \otimes \hat{X}_B \otimes \hat{X}_C|GHZ\rangle = 1$. Hint: For multi-qubit states, operators for one qubit only act on that one just as they would for single qubit states. For example, $\hat{X}_B|000\rangle = |010\rangle$ and $\hat{X}_B|111\rangle = |101\rangle$.

d. **(3 points)** Show that $\langle \hat{X}_A \otimes \hat{Y}_B \otimes \hat{Y}_C \rangle = -1$. By symmetry argue that $\langle \hat{Y}_A \otimes \hat{X}_B \otimes \hat{Y}_C \rangle = \langle \hat{Y}_A \otimes \hat{Y}_B \otimes \hat{X}_C \rangle = -1$ also. Hint: $\hat{Y}_C|000\rangle = i|001\rangle$ and $\hat{Y}_C|111\rangle = -i|110\rangle$.

e. **(5 points)** In a local hidden variable theory, each measurement holds pre-assigned values $x_i, y_i \in \{\pm 1\}$, where $i = A, B, C$. Here, the expectation values from parts (c) and (d) need not be expectations. They give us:

$$x_A x_B x_C = 1, \qquad x_A y_B y_C = -1, \qquad y_A x_B y_C = -1, \qquad y_A y_B x_C = -1.$$

Multiply the latter three equations together and use the fact that $y_i^2 = 1$ to show that local hidden variable theory leads to a logical contradiction with $x_A x_B x_C = 1$, ruling out local realism without the use of any statistics.

Question 3. Superdense Coding (15 points)

Superdense coding allows Alice to send two classical bits to Bob by transmitting a single qubit, using a shared entangled pair. Alice and Bob share the Bell state

$$|\phi^+\rangle_{AB} = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$$

Alice holds qubit A and Bob holds qubit B. Alice wishes to send one of four two-bit messages: 00,01,10,11. She encodes her message by applying the following to her qubit A:

$$00: \hat{I}, \qquad 01: \hat{X}, \qquad 10: \hat{Z}, \qquad 11: \hat{X}\hat{Z}$$

Note that when applying an operator on one qubit, the other qubit remains unchanged, i.e., it is acted upon by the identity. Suppose we want to apply $\hat{X}$ on qubit A, the $\hat{I}$ on qubit B is implicitly assumed: $\hat{X}_A|\psi_{AB}\rangle = (\hat{X}_A \otimes \hat{I}_B)(|\psi_A\rangle \otimes |\psi_B\rangle) = \hat{X}_A|\psi_A\rangle \otimes \hat{I}_B|\psi_B\rangle = \hat{X}_A|\psi_A\rangle \otimes |\psi_B\rangle$.

a. **(8 points)** Show that Alice's encoding scheme produces four orthogonal states (the Bell states). You don't have to show that they are orthogonal.

b. **(2 points)** Alice sends Bob qubit A, who now holds both qubits. Bob performs two-qubit Bell measurement. Can Bob distinguish between all four states? If so, why?

c. **(2 points)** Suppose Eve intercepts qubit A in transit before it reaches Bob. Why can't she determine which message Alice sent from that qubit alone?

d. **(3 points)** Compare superdense coding with quantum teleportation by completing the following table:

Protocol	Quantum Resource	Channel	Information Transmitted
Quantum Teleportation	1 Bell state	2 classical bits	1 qubit
Superdense Coding	1 Bell state	?	?